

# Standard Data Products

## APPN Aerial SOP — Locked revision 1.02

### Standard Data Products

This page provides an overview of the **standard data products** generated by APPN UAV processing pipelines, including typical spatial resolution, file formats, and software compatibility. By default, products are delivered in **open or widely supported formats** that can be accessed using **freely available tools**.

For the pipelines that generate these products, see Processing Pipelines.

For additional information about all GOBI and CALVIS products see the Gryfn Documentation

### Guiding principles

- Products are generated using standardised, documented pipelines to ensure consistency across platforms and sites.
- Output formats prioritise accessibility using freely available and open-source software.
- Spatial resolution and product definitions are explicit and traceable.
- Products are intended to be suitable for downstream analysis, comparison, and reuse.

### Standard LiDAR-derived products

| Product                           | Description                                                 | Typical resolution   | Typical file size ( <i>indicative</i> )                              | Format         | Software compatibility   |
|-----------------------------------|-------------------------------------------------------------|----------------------|----------------------------------------------------------------------|----------------|--------------------------|
| LiDAR Digital Surface Model (DSM) | Surface model representing canopy and above-ground features | 8 cm                 | Derived from LiDAR point cloud (see below)                           | GeoTIFF (.tif) | QGIS, ArcGIS, GDAL       |
| LiDAR Digital Terrain Model (DTM) | Ground surface model with vegetation removed                | 1 m                  | Derived from LiDAR point cloud (see below)                           | GeoTIFF (.tif) | QGIS, ArcGIS, GDAL       |
| Combined LiDAR Point Cloud        | Aggregated and filtered point cloud                         | Native point spacing | ~110–175 MB per minute of flight (first & last files may be smaller) | LAS (.las)     | QGIS, CloudCompare, PDAL |

*Typical LiDAR file sizes are based on raw LiDAR logging rates documented in the CALViS and GOBI SOPs; DSM and DTM sizes scale with survey area and output resolution.*

### Standard hyperspectral products

| Product          | Description                                          | Typical resolution | Typical file size ( <i>indicative</i> )         | Format             | Software compatibility |
|------------------|------------------------------------------------------|--------------------|-------------------------------------------------|--------------------|------------------------|
| VNIR Orthomosaic | Radiometrically calibrated VNIR hyperspectral mosaic | ~4 cm (GSD-based)  | Several GB per flight (even for short missions) | ENVI (.bin + .hdr) | QGIS, ENVI, SNAP       |
| SWIR Orthomosaic | Radiometrically calibrated SWIR hyperspectral mosaic | ~4 cm (GSD-based)  | Several GB per flight (even for short missions) | ENVI (.bin + .hdr) | QGIS, ENVI, SNAP       |

*Hyperspectral data volumes scale strongly with mission duration, frame period, and number of spectral bands. SOPs note that even short flights routinely generate multi-GB datasets.*

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## Standard RGB products

| Product         | Description                               | Typical resolution          | Typical file size ( <i>indicative</i> )           | Format         | Software compatibility |
|-----------------|-------------------------------------------|-----------------------------|---------------------------------------------------|----------------|------------------------|
| RGB Orthomosaic | High-resolution RGB orthorectified mosaic | Use Gryfn flight calculator | ~30–70 MB per image (image every ~2 s by default) | GeoTIFF (.tif) | QGIS, ArcGIS, GDAL     |